

[← Go Back](#)

Hardware

[Connecting Nano 33 BLE Devices over Bluetooth®](#)

[Controlling RGB LED Through Bluetooth®](#)

[Nano 33 BLE Sense User Manual](#)

[Nano 33 BLE Sense Community Projects](#)

[Edge Impulse with the Nano 33 BLE Sense](#)

[Gesture Recognition with the Nano 33 BLE Sense](#)

[Get Started With Machine Learning on Arduino](#)

[Getting Started with OpenMV](#)

[Reading Temperature & Humidity on Nano 33 BLE Sense](#)

[Connecting Two Nano 33 BLE Sense Boards Through I2C](#)

[Accessing Accelerometer Data on Nano 33 BLE Sense](#)

[Accessing Gyroscope Data on Nano 33 BLE Sense](#)

[Accessing Magnetometer Data on Nano 33 BLE Sense](#)

[Debugging with Lauterbach TRACE32 GDB Front-End Debugger](#)

[Controlling the On-Board RGB LED with Microphone](#)

[Proximity Detection with the Nano 33 BLE Sense](#)

[Detecting Colors with the Nano 33 BLE Sense Board](#)

Home / Hardware / Nano 33 BLE Sense / **Reading Temperature & Humidity on Nano 33 BLE Sense**



This tutorial refers to a product that has reached its end-of-life status.

Reading Temperature & Humidity on Nano 33 BLE Sense

Learn how to measure and print out the humidity and temperature values of your surroundings using the Nano 33 BLE Sense

Author · Jose Garcia · Last revision · 07/17/2024

In this tutorial we will use an **Arduino Nano 33 BLE Sense** board to measure and print out the humidity and temperature values of your surroundings, made possible by the embedded **HTS221** sensor.

Goals

The goals of this project are:

Learn how to output raw sensor data from the Arduino Nano 33 BLE Sense.

Use the HTS221 library.
Print temperature and

ON THIS PAGE

Goals

Hardware & Software Needed

The HTS221 Sensor

Creating the Program

Testing It Out +

Conclusion

Serial Monitor when they are within a certain range.

Create your own temperature and humidity monitor.

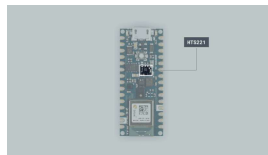
Hardware & Software Needed

Arduino Nano 33 BLE Sense.

This project uses no external sensors or components.

In this tutorial we will use the [Arduino Create Cloud Editor](#) to program the board.

The HTS221 Sensor



The HTS221 sensor.

Temperature sensors are components that convert physical temperature into digital data. Likewise, humidity sensors are able to measure atmospheric moisture levels and translate that into electrical signal. As such, temperature and humidity sensors are essential for environmental monitoring especially in and around sensitive electronic equipment.

The HTS221 is an ultra-compact sensor for relative humidity and temperature. We will use the I2C protocol to communicate with the sensor and get data from it. The sensor's range of different values are the following:

Humidity accuracy: $\pm 3.5\%$
rH, 20 to +80% rH

Humidity range: 0 to 100 %

Temperature accuracy: $\pm 0.5\text{ }^{\circ}\text{C}$, 15 to +40 $^{\circ}\text{C}$

Temperature range: -40 to 120 $^{\circ}\text{C}$

These types of sensors are used more than you would think and are found in various everyday objects!

Some of the useful ways they are used are seen in the following applications:

Air conditioning, heating and ventilation.

Air humidifiers.

Refrigerators.

Wearable devices.

Smart home automation.

Industrial automation.

Respiratory equipment.

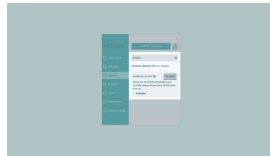
Asset and goods tracking.

If you want to read more about the HTS221 sensor you can take a look at the [datasheet](#).

Creating the Program

1. Setting up

the **Libraries** tab, search for the **HTS221** library, then in **Examples**, open the **ReadSensors** example. Once the sketch is open, rename it as **Temp_Humidity**.



Finding the library in the Cloud Editor.

2. Connecting the board

Now, connect the Arduino Nano 33 BLE Sense to the computer to check that the Cloud Editor recognises it, if so, the board and port should appear as shown in the image. If they don't appear, follow the [instructions](#) to install the plugin that will allow the Editor to recognise your board.



Selecting the board.

3. Printing temperature and humidity values

Now we will need to modify the code of the example in order to print the temperature and humidity values only when 0,5°C degree has changed or the humidity has changed at least 1%. To do so, let's initialize the following variables before the

```
1 float old_temp = 0;
2 float old_hum = 0;
```

In the `setup()`, the `HTW.begin()` function inside an `if` statement will print a message, as a string, in case the sensor has not been properly initialized.

Then, in the `loop()`, let's modify the example code by adding the following portion of code after the `HTS.readTemperature()` and `HTS.readHumidity()` functions respectively.

```
1 // check if the range
2 // and if the range
3 if (abs(old_temp -
4 {
5     old_temp = temper
6     old_hum = humidit
7     // print each of
8     Serial.print("Tem
9     Serial.print(temp
10    Serial.println("
11    Serial.print("Hum
12    Serial.print(humi
13    Serial.println("
14    Serial.println();
15 }
```

With this part of the code, we will print out the temperature and humidity values when the temperature increases or decreases more than 0,5°C , or, when the humidity values change up or down more than

4. Complete code

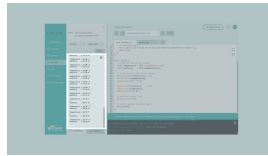
If you choose to skip the code building section, the complete code can be found below:

```
1  /*
2   HTS221 - Read Sens
3
4   This example reads
5   Nano 33 BLE Sense
6   values to the Seri
7
8   The circuit:
9   - Arduino Nano 33
10
11  This example code
12  */
13
14  #include <Arduino_Hi
15
16  float old_temp = 0;
17  float old_hum = 0;
18
19  void setup() {
20    Serial.begin(9600)
21    while (!Serial);
22
23    if (!HTS.begin())
24      Serial.println("
25      while (1);
26  }
27  }
28
```

Testing It Out

After you have successfully verified and uploaded the sketch to the board, open the Serial Monitor from the menu on the left. You will now see the new values printed. If you want to test out whether it is working, you could slightly breathe

humidity, as well as the temperature, levels rise or decrease. The following image shows how the data should be displayed.



Temperature & humidity printed in the Serial Monitor.

Troubleshoot

Sometimes errors occur, if the project is not working as intended there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon.

- If your Arduino board is not recognised, check that the Create plugin is running properly on your computer

- Accidental interruption of cable connection.

Conclusion

In this simple tutorial we learned how to read temperature and humidity values from the **HTS221** sensor using the **HTS221 library**, and how to use the sensor embedded in the Arduino Nano 33 BLE Sense board, to measure and print out humidity and temperature

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